

5 TASTE 100 HERBS: MATERIAL SCARCITY AND LOCAL PLANT KNOWLEDGE IN THE MAO-ERA CAMPAIGN FOR NATIVE PESTICIDES

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In August 1960, in the midst of the worst famine in world history, the Kaifeng Municipal Science and Technology Committee published a reference volume for local production of fertilizers and pesticides (figure 5.1).¹ The book was handwritten and mimeographed on low-grade recycled paper that was soft, gray, fibrous, and speckled with bits of darker pulp or sometimes even pieces of bark or twig. The pages—both the words themselves and the material on which they were inked—spoke to scarcity and resourcefulness. The text served the urgent priority of food production: wild plants, together with every other available resource, would be combined according to knowledge distilled from traditional Chinese doctors and modern chemists, and then applied to crops to boost growth and forestall disease and insect pests. The book's paper, too, followed the principle of making do with what was available: as land once used to grow bamboo for papermaking was converted to food production,² mills turned to inferior sources for their raw materials. In places with severe shortages (including the region around Kaifeng), rural people ate bark, twigs, and some of those same wild plants described in such handbooks, sometimes poisoning themselves in the process. During those years, the lines between food, medicine, poison, and even paper became more blurred than usual.

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1960年8月，在世界历史上最严重的饥荒中，开封市科学技术委员会出版了一本关于当地生产化肥和农药的参考书（图5.1）。¹这本书是手写的，用低档再生纸油印而成，纸质柔软、灰色、多纤维，上面点缀着深色纸浆，有时甚至是树皮或树枝的碎片。书页——无论是文字本身还是印制文字的材料——都昭示着稀缺和足智多谋。文本服务于粮食生产的紧迫优先事项：野生植物与所有其他可用资源相结合，根据从传统中医和现代化学家那里提炼出的知识，然后应用于作物以促进生长并预防疾病和虫害。这本书的纸张也遵循了就地取材的原则：随着曾经用于种植竹子造纸的土地被转用于粮食生产，²造纸厂转而使用劣质原料。在严重短缺的地区（包括开封周边地区），农村人吃树皮、树枝和一些手册中描述的野生植物，有时在此过程中中毒。在那些年里，食物、药品、毒药甚至纸张之间的界限变得比以往任何时候都模糊。

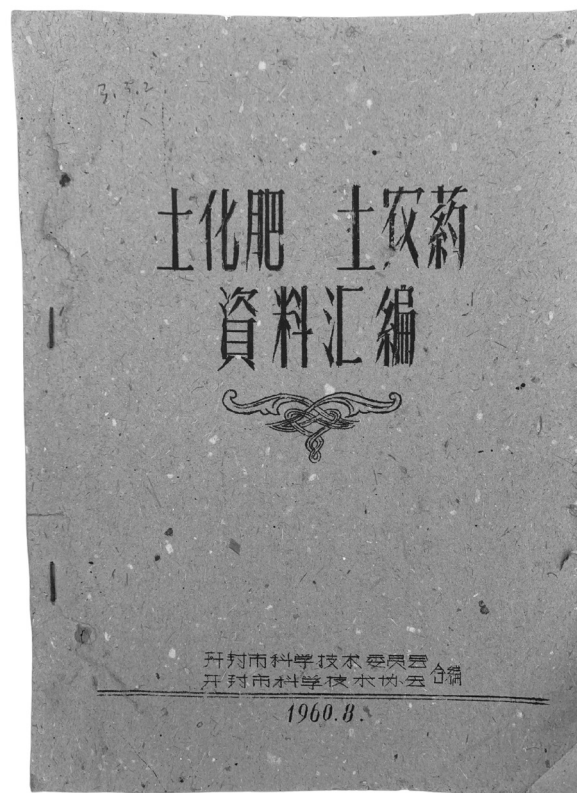


FIGURE 5.1

The cover of a handbook on native fertilizers and pesticides, published in Kaifeng, 1960. Photo by Jessica Johnson; book in author's collection.

This chapter explores the *da gao tunongyao yundong* (大搞土农药运动 mass campaign to manufacture native pesticides), first launched in 1958 to support food production during the Great Leap Forward. The campaign was part of the PRC leadership's effort to overcome material scarcity and transform agriculture along both modern and revolutionary lines. It relied on and celebrated local knowledge of the myriad uses of wild and cultivated plants, systematizing this knowledge for the first time in history to help local governments manufacture botanical, “native” pesticides. Of special importance was local people's knowledge of the medicinal properties of plants, including their traditional classification in terms of *xingwei* (性味 character of taste/smell): bitter, spicy, numbing, acrid, stifling, and sour.

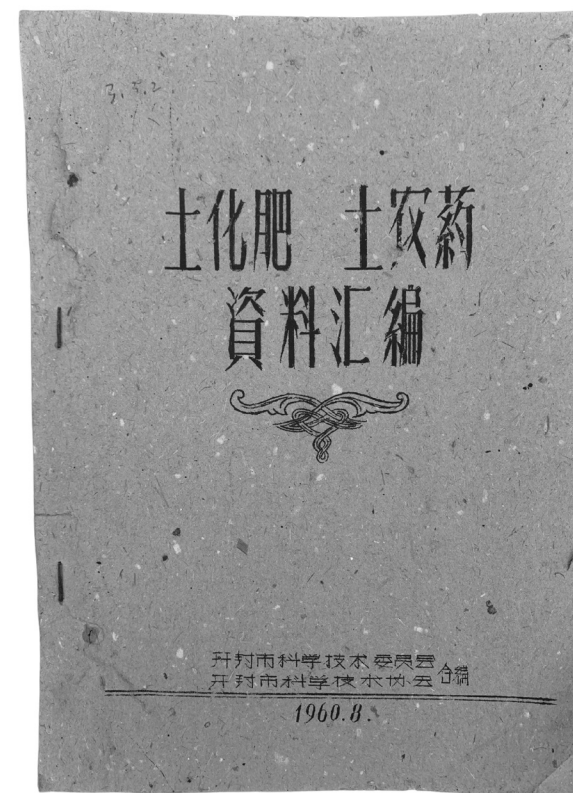


图5.1 1960年在开封出版的《土化肥农药手册》封面。杰西卡·约翰逊摄；作者藏书。

本章探讨了大搞土农药运动（大搞土农药运动 大规模制造土农药运动），该运动于1958年首次发起，旨在大跃进期间支持粮食生产。该运动是中国领导层努力克服物质匮乏，并以现代化和革命化方式改造农业的一部分。它依赖并颂扬了当地对野生和栽培植物无数用途的知识，历史上首次将这些知识系统化，以帮助地方政府制造植物性“土”农药。特别重要的是当地人民对植物药用特性的知识，包括它们在性味（性味 味/嗅觉特征）方面的传统分类：苦、辣、麻、涩、闷和酸。

Inserting this topic into the larger history of foodways in modern China is an unusual move, but it is an important one if we are to grapple conscientiously with the complexities of the Mao era and its relationship to the rest of modern Chinese history. The question of food in Mao-era China usually begins and ends with the question of food availability—or, more to the point, unavailability. Indeed, failing to center the reality of hunger and famine would constitute a grave injustice against the people who suffered from them. However, the overwhelming narrative of deprivation, together with the broader repudiation of radical politics, has prevented recognition of the relevance of the Mao era—and perhaps especially the Great Leap Forward—to the questions animating conversations about food and agriculture in China today. In his study of the culinary traditions of Guangzhou, for example, Jakob Klein has demonstrated that despite the very real food shortages experienced during the Mao era, “attempts to create a socialist food culture not only relied upon but even came to celebrate and solidify preexisting culinary practices and institutions,” the legacies of which can be traced in post-Mao Guangzhou cuisine.³ In this chapter, I similarly argue that the Mao era represents important continuities in a history that spans the pre-1949 and post-1978 periods, but I shift the inquiry from urban to rural society and select a case that highlights other knowledge forms—including agricultural and medicinal—central to the production of food.

The Kaifeng handbook, along with many other materials from the native pesticides campaign, provides an illuminating window onto modern Chinese foodways precisely because it brings into focus the complex relationships among scarcity and abundance; modernization and tradition; technoscience and local knowledge; state and rural society; plants and people; nourishment and poison; and food, agriculture, and medicine. These themes resonate strongly with the concerns of alternative food and agriculture movements today, and they demonstrate compelling continuities in the history of Chinese foodways throughout the modern period, as evidenced by many of the other chapters of this book. The dominant narrative jumps over the discredited Mao era and appeals directly to imperial-era “tradition.” In contrast, this chapter demonstrates the pivotal role played by the Great Leap Forward within the larger history of Chinese foodways, in which state, scientific, and rural actors have sought to address the dilemmas of industrialized agriculture by systematizing local knowledge. In the process, it underscores the need to respect the enormity of what Mao-era historical actors were attempting in

将这个主题纳入中国现代饮食史的宏大叙事中，是一个不同寻常的举动，但如果我们要认真应对毛泽东时代的复杂性及其与中国现代史其余部分的关系，这又是至关重要的一步。毛泽东时代中国的食物问题通常始于也终于食物的可获得性——或者更确切地说，是不可获得性。确实，不以饥饿和饥荒的现实为中心，将构成对遭受这些苦难的人民的严重不公。然而，压倒性的匮乏叙事，加上对激进政治的广泛否定，阻碍了人们认识到毛泽东时代——也许特别是大跃进——与当今中国关于食物和农业的讨论所激发的问题之间的关联性。例如，雅各布·克莱因（Jakob Klein）在他对广州烹饪传统的研究中指出，尽管毛泽东时代经历了非常真实的食物短缺，“创造社会主义饮食文化的尝试不仅依赖于，甚至颂扬并巩固了既有的烹饪实践和制度”，其遗产可以在后毛泽东时代的广州菜中追溯到。³在本章中，我同样认为毛泽东时代代表了跨越1949年前和1978年后的历史中的重要连续性，但我将探究的重点从城市社会转向农村社会，并选择了一个突出其他知识形式——包括农业和医药——对食物生产至关重要的案例。

《开封手册》以及许多其他来自本土农药运动的材料，为我们提供了一个了解现代中国饮食方式的窗口，因为它精确地聚焦于稀缺与丰饶、现代化与传统、科技与地方知识、国家与乡村社会、植物与人类、营养与毒药以及食物、农业与医学之间复杂的关系。这些主题与当今另类食物和农业运动的关注点强烈共鸣，它们展示了中国饮食方式在整个现代时期引人注目的连续性，正如本书许多其他章节所证明的那样。主流叙事跳过了声名狼藉的毛泽东时代，直接诉诸帝国时代的“传统”。相比之下，本章展示了大跃进在中国饮食方式的宏大历史中所扮演的关键角色，其中国家、科学和乡村行动者通过系统化地方知识来解决工业化农业的困境。在此过程中，它强调了尊重毛泽东时代历史行动者所尝试的巨大努力的必要性。

the transformation of food production, to recognize the consequences for rural people struggling to satisfy their most fundamental food needs, and to do justice to the positive role played by diverse actors in mobilizing local plant knowledge.

SCARCITY

The native pesticides campaign was one small part of Chairman Mao's effort to create full-fledged communism within the span of just a few years. Marxists had typically foreseen a long process of development under state socialism before society would achieve the ideal, "from each according to their ability, to each according to their needs"; in 1958, with the launch of the Great Leap Forward, Mao proposed that the revolutionary spirit of the masses could propel China to that stage far more quickly. But spirit and organization alone would not be enough: the Leap required the transformation of material production. This meant the rapid development of industry, which relied on a massive increase in food (and especially grain) production—enough not only to feed China's laborers but also to send to the Soviet Union in exchange for machinery, oil, and other industrial inputs.

Increasing food production required the rapid modernization of many aspects of agriculture. Mao specifically itemized eight necessary areas for improvement, of which crop protection (controlling plant diseases and insect pests) constituted one. But the same scarcity that China's leaders sought to overcome with the Great Leap Forward posed a limit on crop protection as on so much else: China lacked the quantities of chemical pesticides that developed countries were increasingly employing to protect their harvests, and it could not rely on foreign countries to supply them.

The pressure to accelerate agricultural production led to badly conceived technological changes, outrageous harvest projections, downright lies about surpluses, and far greater extraction than many rural communities could bear. At the same time, the Leap's utopian vision led to the establishment of canteens where commune members could eat to excess without payment—a claim to abundance that belied continued struggles with scarcity and would appear tragically short-sighted when famine set in.⁴ These are the most well-known aspects of the Great Leap Forward. But there was far more to it than that. To achieve targets and demonstrate revolutionary zeal, people around

粮食生产的转型，认识到其对
努力满足最基本粮食需求的农村人口造成的影响，并
公正对待不同参与者在调动当地

植物知识方面发挥的积极作用。

稀缺性

本土农药运动是毛主席在短短几年内建立全面共产主义的努力的一小部分。马克思主义者通常预见到，在国家社会主义下，社会要经历一个漫长的发展过程，才能达到“各尽所能，各取所需”的理想境界；1958年，随着大跃进的启动，毛主席提出，群众的革命精神可以更快地推动中国达到这一阶段。但仅仅依靠精神和组织是不够的：大跃进需要物质生产的转型。这意味着工业的快速发展，而这又依赖于粮食（尤其是谷物）产量的巨大增长——不仅要足以养活中国的劳动力，还要能够输送给苏联，以换取机械、石油和其他工业投入。

增加粮食产量需要农业许多方面的快速现代化。毛主席特别列举了八个需要改进的领域，其中作物保护（控制植物病害和虫害）是其中之一。但中国领导人试图通过大跃进克服的稀缺性，对作物保护以及其他许多方面都构成了限制：中国缺乏发达国家越来越多地用于保护其收成的化学农药，而且它不能依赖外国供应。

加速农业生产的压力导致了构思拙劣的技术变革、离谱的产量预测、公然谎报盈余，以及远远超出许多农村社区承受能力的过度开采。与此同时，“大跃进”的乌托邦愿景导致了公共食堂的建立，社员可以在那里免费吃到饱——这种对富足的宣称掩盖了持续的稀缺困境，并在饥荒来临时显得极具悲剧性的短视。⁴ 这些都是大跃进最广为人知的方面。但其影响远不止于此。为了实现目标和展示革命热情，各地的人们

the country also tapped existing knowledge and developed innovative strategies to maximize production with limited resources.

Local knowledge of wild plants—seemingly a limitless resource requiring no inputs—presented a particularly fruitful area to explore. Depending on the species, wild plants supplied fiber (for clothing and paper), oil (including for food, soap, candles, and paint), starch (for food), tannin (for processing leather), rubber, resin, pesticides, and of course medicine. A 1958 article in *Bulletin of Biology* detailed all of the above uses along with the plants that could supply them and their availability; it covered pesticides in the same section along with medicinal herbs. Indeed, looked at from some angles, pesticides were a kind of medicine: the Chinese term for pesticides, “*nongyao*” (农药 literally “agricultural medicine”), includes treatments for plant diseases and sometimes also fertilizers. With respect to using wild starchy plants for food, the article noted, “If we can make comprehensive use [*chongfen liyong* 充分利用] of wild starches, not only can we increase peasant income, but we can also save the country much grain.” Wild plants could supply substitutes for maltose, tofu, bean thread noodles, lotus root paste, processed cakes, and other foodstuffs. For example, according to the article, China had just that year substituted 100,000 tons of acorns for 70,000 tons of food crops, in the process raising peasant income by 12 million yuan.⁵

In May 1958, an article in *Insect Knowledge* noted that the national supply of chemical pesticide for that year was 480,000 tons, and they had 3.1 million pesticide sprayers; given the “high tide” of the Great Leap Forward these were not enough to satisfy demand. Hence the call for every area to adopt the policy of “simultaneously pursuing native [*tu* 土] and foreign [*yang* 洋] methods to go big in native insecticides.”⁶ Over the course of 1958, the participating localities reported the establishment of more than 2.4 million native pesticide factories, manufacturing more than five hundred types of pesticides mixed into more than three hundred compound sprays, and applying 16 million tons of spray to the fields.⁷ The scarcity of chemical pesticides was the incessant drumbeat driving the native pesticide campaign, and this scarcity was portrayed as an opening to develop native self-reliance.

In truth, the supply of wild plants was by no means unlimited. Rather, the call to harvest wild plants for use as native pesticides quickly ran up against other priorities. As one article from June 1958 noted, some of the plants were needed more urgently for human medicine; some were needed for oil, fiber,

该国还利用现有知识并制定了创新战略，以有限的资源实现生产最大化。

关于野生植物的本土知识——这似乎是一种无限的资源，不需要任何投入——提供了一个特别有益的探索领域。根据物种的不同，野生植物提供纤维（用于服装和纸张）、油（包括用于食物、肥皂、蜡烛和油漆）、淀粉（用于食物）、单宁（用于加工皮革）、橡胶、树脂、杀虫剂，当然还有药物。1958年发表在《生物学通报》上的一篇文章详细介绍了上述所有用途以及可以提供它们的植物及其可用性；它在同一部分中涵盖了杀虫剂和药草。事实上，从某些角度来看，杀虫剂是一种药物：中文中“*nongyao*”（农药，字面意思是“农业药物”）一词包括植物病害的治疗，有时也包括肥料。关于使用野生淀粉植物作为食物，文章指出，“如果我们能[*chongfen liyong* 充分利用]野生淀粉，不仅可以增加农民收入，还可以为国家节省大量粮食。”野生植物可以替代麦芽糖、豆腐、粉丝、藕粉、加工糕点和其他食品。例如，根据文章，中国当年就用10万吨橡子替代了7万吨粮食作物，在此过程中使农民收入增加了1200万元。⁵

1958年5月，《昆虫知识》杂志刊登的一篇文章指出，当年全国化学农药供应量为48万吨，拥有310万台农药喷雾器；鉴于大跃进的“高潮”，这些不足以满足需求。因此，文章呼吁各地采取“土洋结合”的政策，大力发展土农药。⁶ 1958年，各地报告称建立了240多万家土农药厂，生产了500多种农药，混合成300多种复合喷剂，向农田喷洒了1600万吨喷剂。⁷ 化学农药的稀缺性是推动土农药运动的持续动力，这种稀缺性被描绘成发展本土自力更生的契机。

事实上，野生植物的供应绝非无限。相反，号召采集野生植物用作土农药很快就与其他优先事项相冲突。正如1958年6月的一篇文章所指出的，有些植物更迫切地需要用于人类医药；有些则需要用于提炼油、纤维，

or starch; and others were valuable as timber. Arguments had already begun to erupt as to how the different needs could be balanced. In fact, figuring out how such resources could be allocated rationally was itself a question that required scientific research.⁸ A publication from Da County, Sichuan—a frontrunner in the campaign—similarly highlighted the supply problem and noted that the county had begun cultivating plants useful for native pesticides in the “ten margins”: that is, the edges of fields, mountains, ditches, rivers, homes, graves, and other places where more plantings could be squeezed in.⁹ And so, what had been trumpeted as a limitless bounty pointed instead, again, to scarcity.

Moreover, and as we will see below, the food substitution project that seemed so positive in 1958 would in a few years take on a much less pleasant flavor. Still, we would be wrong to dismiss these ideas out of hand: exploring new ways of using available resources was a fundamentally rational response to the crucial problem of scarcity, and it remains a touchstone for resource conservation and management in China to this day. Accounting for such efforts and recognizing the magnitude and persistence of the development challenge China has faced helps us move away from a smug dismissal of this painful history, and toward an empathetic recognition of complexity, along with an acknowledgment of the many shared priorities of the Great Leap era and today.

KNOWLEDGE

The scarcity of chemical pesticides presented an immensely practical reason to ramp up a policy orientation that, ever since the early 1940s in the revolutionary base area of Yan'an, had already enjoyed great ideological significance for the CCP: the development of *tu* (土 local or native) science to serve as a complement to *yang* (洋 foreign) science, as in the slogan “*tuyang jiehe*” (土洋结合 combine *tu* and *yang*). Thus, in the campaign for native pesticides, *tu* referred not only to the pesticides themselves but also to the type of science being practiced. In this respect, it is important to note the multivalent associations of *tu* and *yang*, beyond those implied by the similar pair *zhong* (中 China) and *xi* (西 the West). While both binaries point to the differences between “Chinese” and “Western,” *tu* (literally “soil”) further implies rural and mass-based, in contrast with *yang* (literally “ocean”), which implies academic and elite.¹⁰ To those familiar with contemporary Chinese food culture,

或淀粉；另一些则作为木材具有价值。关于如何平衡不同需求争论已经爆发。事实上，弄清楚如何合理分配这些资源本身就是一个需要科学研究的问题。⁸ 四川大县（这场运动的先行者）的一份出版物也强调了供应问题，并指出该县已经开始在“十边”种植对本地农药有用的植物：即田边、山边、沟边、河边、房边、墓边以及其他可以挤入更多种植的地方。⁹ 因此，曾经被吹嘘为无限的恩惠，却再次指向了稀缺性。

此外，正如我们将在下文看到的那样，1958年看起来如此积极的食物替代项目，几年后会变得不那么令人愉快。尽管如此，我们如果立刻否定这些想法是错误的：探索利用现有资源的新方法是对稀缺性这一关键问题的根本性理性回应，它至今仍是资源保护和管理的试金石。解释这些努力并认识到中国所面临发展挑战的巨大性和持续性，有助于我们摆脱对这段痛苦历史的自鸣得意，转而同情地认识其复杂性，并承认大跃进时期和今天有许多共同的优先事项。

知识

化学农药的稀缺性提供了一个极其现实的理由，促使人们加强一项政策导向，这项政策自20世纪40年代初在延安革命根据地以来，对中国共产党而言已经具有重要的思想意义：发展*tu*（土，本土的或原生的）科学，作为*yang*（洋，外国的）科学的补充，正如“*tuyang jiehe*”（土洋结合，土洋结合）的口号。因此，在推广本土农药的运动中，*tu*不仅指农药本身，也指所实践的科学类型。在这方面，值得注意的是*tu*和*yang*的多重关联，超越了与*zhong*（中，中国）和*xi*（西，西方）这对相似词所暗示的关联。虽然这两对二元对立都指向“中国”和“西方”之间的差异，但*tu*（字面意思是“土壤”）进一步暗示了乡村和群众基础，与*yang*（字面意思是“海洋”）形成对比，后者暗示了学术和精英。¹⁰对于熟悉当代中国饮食文化的人来说，

zhong may be more familiar in discourse on cuisine, while *tu* appears in more “earthy” contexts of locally raised heritage breeds—as in *tu zhu* (土猪 native pigs). *Tu* science was thus not just Chinese but also local, earthy, frugal, and peasant-based science.

Tu science celebrated making do with locally available materials, and this, of course, was what the native pesticide campaign was all about. But for the frontrunners of the campaign in particular, another aspect of *tu* science was also essential: local knowledge. The national Ministry of Agriculture did not release its encyclopedia of native pesticides until September 1958—months into the campaign. Until then, cadres tasked with developing native pesticides had to consult more accessible authorities, and they quite logically turned first to traditionally trained doctors, pharmacists, and veterinarians. The treatment of crops that fed people had historically been linked to the treatment of people themselves, and wild plants were the foundation of that knowledge. For example, rice blast disease in Chinese is *dao wenyi* (稻瘟病 rice febrile epidemic) or *dao rebing* (稻热病 rice hot disease), suggesting a kinship with the human ailments known in traditional Chinese medicine as *wenbing* (温病 warm-factor disorders).¹¹ As we will see below, herbs useful in treating warm-factor disorders in people are also used in treating rice blast disease. Some of the traditional medical experts consulted by the cadres reported having experience treating crops—for example, using *tonggen* (桐根 *Paulownia* roots) to prevent damage from cabbage moths, the legume *Milletia pachycarpa* to combat cutworms, and *Stemona* vines to control corn borers. They also knew the proper preparation methods and the timing of applications.¹² One article published in *Insect Knowledge* pointed out that even “old monks” had knowledge of wild plants that should be tapped for the campaign.¹³

Materials on native pesticides often cited the long history of their use in China. As one article in *Insect Knowledge* declared, “Several thousand years ago, in the process of their productive labor, our ancestors unearthed and used abundant native pesticides. They accumulated a great deal of experience, which in the current plan of combining *tu* and *yang* constitutes an extremely valuable scientific heritage.”¹⁴ One of the most powerful botanical insecticides promoted during the campaign was rotenone, which was found in chicken-blood vine among other legumes of southern China, and which one 1958 article noted with pride was first discovered by “Chinese laboring people” and first recorded in 1557 in the “great Chinese scientist Li Shizhen’s *Materia Medica*.” The plants themselves were originally used as a piscicide

中在烹饪话语中可能更常见，而土则更多地出现在本地饲养的传统品种的“乡土”语境中——例如土猪（土猪）。因此，土科学不仅是中国本土的，而且是地方的、乡土的、节俭的、以农民为基础的科学。

土科学推崇就地取材，这当然正是本土农药运动的全部意义所在。但对于这场运动的先行者来说，土科学的另一个方面也至关重要：地方知识。国家农业部直到1958年9月——运动开始数月后——才发布其本土农药百科全书。在此之前，负责开发本土农药的干部必须咨询更容易接触到的权威人士，他们很自然地首先求助于受过传统训练的医生、药剂师和兽医。对人类赖以生存的作物的治疗历来与对人类自身的治疗联系在一起，而野生植物是这种知识的基础。例如，水稻稻瘟病在中文中是稻瘟病(稻瘟病水稻热性流行病)或稻热病(稻热病水稻热病)，这表明它与中医中被称为温病(温病温邪病)的人类疾病有亲缘关系。¹¹正如我们将在下文看到的，用于治疗人类温邪病的草药也用于治疗水稻稻瘟病。一些干部咨询的传统医学专家报告说有治疗作物的经验——例如，使用桐根(桐根 泡桐根)来防止菜蛾的损害，豆科植物密花豆来防治地老虎，以及百部藤来控制玉米螟。他们还知道正确的制备方法和施用时间。¹²《昆虫知识》上发表的一篇文章指出，即使是“老和尚”也拥有野生植物知识，应该为这场运动所用。

13

关于本土农药的材料经常引用其在中国悠久的历史。正如《昆虫知识》一篇文章宣称：“几千年前，我们的祖先在生产劳动过程中，发掘并使用了丰富的本土农药。他们积累了大量经验，这在当前土洋结合的计划中构成了极其宝贵的科学遗产。”¹⁴在这次运动中推广的最有效的植物杀虫剂之一是鱼藤酮，它存在于鸡血藤和中国南方其他豆科植物中。1958年的一篇文章自豪地指出，鱼藤酮是“中国劳动人民”首次发现的，并于1557年首次记载于“中国伟大科学家李时珍的《本草纲目》”中。这些植物本身最初被用作鱼藤。

(they easily kill fish that can then be consumed by humans without apparent ill effects); the chemical rotenone had subsequently been isolated and employed worldwide as an insecticide.¹⁵

In fact, however, there was no focused, systematic attention to botanical pesticides as a field of knowledge, at least in the written record, during the imperial era. The famous and comprehensive series *Science and Civilisation in China* devotes forty-eight pages to the subject but acknowledges that the evidence is “scattered.” The majority of examples it considers from imperial-era texts do not actually refer to agricultural uses of the herbs at all, but rather to killing lice, fleas, intestinal worms, and other human afflictions. Indeed, one of the sources it relies on most heavily is a volume published in 1959 by the Joint Office for Scientific Research on Native Pesticides, discussed in detail in the next section.¹⁶

Nonetheless, the connection to agricultural heritage created something of a mystique around the use of wild plants to produce native pesticides, and sometimes authors drew on classical language to convey this sentiment. Evoking the Song dynasty poet Su Shi’s Daoist-inflected *Red Cliff Rhapsody*, one article opined, “China’s native pesticide resources are extremely abundant: we can take them without end, use them without exhausting them (*qu zhi bu jin, yong zhi bu jie* 取之不尽·用之不竭).” The author further called on the legendary ruler Shennong, whose most celebrated contribution to Chinese civilization was the identification of medicinal plants: “So the masses say: Everyone acts as Shennong, going everywhere and tasting one hundred herbs, changing the useless into the useful, changing poisonous plants into agricultural medicine [i.e., pesticides].”¹⁷ Such references are familiar fare in agricultural heritage discourse today, as found in Bray’s chapter in this volume (chapter 1): their significance in Great Leap-era discourse bears remembering.

Traditional medicinal knowledge was not limited to an inventory of useful herbs: it also provided principles to help determine which to combine and which to separate. According to some sources, it was essential to identify the classification of the herb in terms of the *xingwei* (性味 character of taste/smell) of traditional medicine, specifically whether a given plant was bitter, spicy, numbing, astringent, stifling, or sour. On the basis of these classifications, herbs would be separated for storage. While it was fine to store bitter herbs together with spicy herbs, numbing herbs should be kept separate from spicy ones.¹⁸

(它们很容易杀死鱼类，而这些鱼类在被人食用后似乎没有不良影响)；化学物质鱼藤酮随后被分离出来，并在世界范围内用作杀虫剂。¹⁵

然而，事实上，在帝国时期，至少在书面记录中，并没有将植物农药作为一个知识领域进行集中、系统的关注。著名而全面的系列丛书《中国科学技术史》用四十八页的篇幅专门讨论了这一主题，但承认证据是“零散的”。它所考察的帝国时期文献中的大多数例子实际上根本没有提及草药的农业用途，而是指杀死虱子、跳蚤、肠道蠕虫和其他人类疾病。事实上，它最依赖的资料来源之一是1959年由中草药农药科学研究联合办公室出版的一本书，这将在下一节详细讨论。¹⁶

尽管如此，与农业遗产的联系在利用野生植物生产天然农药方面营造了一种神秘感，有时作者会借用古典语言来表达这种情感。一篇文章援引宋代诗人苏轼道家风格的《赤壁赋》写道：“中国的天然农药资源极其丰富：我们可以取之不尽，用之不竭 (*qu zhi bu jin, yong zhi bu jie* 取之不尽·用之不竭)。”作者还提到了传说中的统治者神农，他对中华文明最著名的贡献是鉴定药用植物：“所以大众说：人人都是神农，走遍天下，尝遍百草，化无用为有用，化毒草为农药[即农药]。”¹⁷ 这种引用在今天的农业遗产论述中很常见，如布雷在本卷中的章节（第一章）所述：它们在大跃进时期的论述中的重要性值得铭记。

传统医药知识不仅限于有用的草药清单：它还提供了有助于确定哪些草药可以组合、哪些草药应该分开的原则。根据一些资料，识别草药的分类至关重要，这要根据传统医药的性味（性味，即味道/气味的特征），特别是某种植物是苦、辣、麻、涩、闷还是酸。根据这些分类，草药将被分开储存。虽然苦味草药可以和辛辣草药一起储存，但麻味草药应该与辛辣草药分开存放。¹⁸

People also recommended employing the principle of *shiba fan yao* (十八反药 eighteen conflicting medicines) to identify toxic combinations—for example, onions and honey. While this principle had originally been developed to help people avoid ingesting *fan* (反 conflicting) foods and medicines whose combination would produce toxins in the body, the knowledge could also be turned around and used the other way. The afterword to a native pesticide handbook published in Da County detailed which combinations increased toxicity and thus effectiveness (e.g., numbing plus bitter, or spicy plus bitter), which decreased it (e.g., numbing plus spicy—the famous *mala* combination characteristic of Sichuan cuisine), and which had no effect (e.g., bitter plus stifling).¹⁹

An article by the Da County Department of Commerce explained that to increase moistness or suspension in a compound, they added material based on the *xingwei* of each herb. “For example, add 2–3 percent honey locust to spicy herbs, 4–5 percent chinaberry to bitter herbs, etc.” (The “etc.” here implied that readers should know how to apply this principle to herbs with other characteristics, suggesting an assumption that such medicinal knowledge was indeed widespread among rural people.) The authors further noted the importance of an herb’s *xingwei* in the manufacturing process: some herbs (especially those with numbing *xingwei*) were sensitive to heat. So, for example, when preparing garlic for use against rice blast disease, it was important to use cool water.²⁰

In the spirit of *tu* science, some texts suggested that ordinary peasants, not just doctors or pharmacists, possessed valuable knowledge about wild plants. One article in *Insect Knowledge* explained, “Our ancestors began using native pesticides in ancient times and accumulated thousands of years of rich experience that they have passed down to the people. All peasants have experience identifying, collecting, and using poisonous plants. They don’t need equipment; all they need is to distinguish the *xingwei* (sour, spicy, bitter, astringent, numbing, stifling) to determine toxicity.” The author went on to cite the example of an “old peasant” in Anhui Province named Feng Ruizhai who used an herb called *mao’er yan* (猫儿眼 or 甘遂 “kitten eyes” or *gansui*) to control the disease known as wheat rust, with an efficacy of 80–100 percent. “These kinds of people are everywhere,” the article emphasized.²¹

Another article highlighted an example from the Dong nationality region of Baise in Guangxi, where commune members created a compound

人们还建议采用十八反药(十八反药十八种相克药物)的原则来识别有毒组合——例如，洋葱和蜂蜜。虽然这一原则最初是为了帮助人们避免摄入反(反相克)食物和药物，因为它们的组合会在体内产生毒素，但这些知识也可以反过来利用。大县出版的一本本土农药手册的后记详细说明了哪些组合会增加毒性从而提高药效(例如，麻木加苦味，或辛辣加苦味)，哪些会降低药效(例如，麻木加辛辣——著名的麻辣组合，四川菜的特色)，以及哪些没有效果(例如，苦味加闷)。¹⁹

大县商务局的一篇文章解释说，为了增加化合物的湿润度或悬浮性，他们根据每种草药的性味添加材料。“例如，在辛辣草药中加入2-3%的皂荚，在苦味草药中加入4-5%的苦楝等。”(这里的“等”字暗示读者应该知道如何将此原则应用于具有其他特性的草药，这表明人们认为这种药用知识在农村地区确实很普遍。)作者进一步指出草药的性味在制造过程中的重要性：有些草药(特别是那些具有麻木性味的草药)对热敏感。因此，例如，在制备用于防治稻瘟病的蒜时，使用冷水很重要。²⁰

本着乡土科学的精神，一些文章指出，不仅医生或药剂师，普通农民也掌握着关于野生植物的宝贵知识。《昆虫知识》中的一篇文章解释道：“我们的祖先在远古时代就开始使用本土农药，积累了数千年的丰富经验并传承给后人。所有农民都有识别、收集和使用有毒植物的经验。他们不需要设备；他们所需要的就是辨别性味(酸、辣、苦、涩、麻、闷)来判断毒性。”作者接着引用了安徽省一位名叫冯瑞斋的“老农民”的例子，他使用一种名为猫儿眼(猫儿眼或甘遂)的草药来防治小麦锈病，效果达到80-100%。文章强调：“这种人随处可见。”²¹

另一篇文章重点介绍了广西百色侬族地区的一个例子，当地社员研制出一种复方制剂

made of *chafu* (茶麸 tea bran), a plant known in the Dong language as *guoxin ye* (果心叶 fruit-heart leaves), ashes, and an herb known as “chicken blood” (known elsewhere as “chicken-blood vine” and by numerous other names); it was found to be 95 percent effective against rice borers. The article underscored, “There are similar examples all over the country.”²²

The mobilization of rural people and the application of traditional medical knowledge (framed as the accumulated wisdom of laboring Chinese people over the centuries) in local pesticide factories: this was quintessential *tu* science. But the native pesticide campaign was expected to create a “material basis for the uniting of *tu* and *yang*” and so “open a new road for technological revolution.”²³ Where *yang* science came in was the systematic organization and dissemination of local experiences, the chemical analysis of herbs, and the testing of the pesticides by professional researchers.

The Joint Office for Scientific Research on Native Pesticides, founded in 1958, conducted research directly on the pesticides and their targets. So as not to lose time over the winter and spring, they bred insects in the lab and tested the pesticides there—recognizing that the conditions were different from those in the field but maintaining nonetheless the usefulness of the information produced. They tested 404 types of native pesticides on mosquito larvae. In each case they tested a tincture made with water and another made with alcohol, and they also compared the same species collected from different locations, though the reasons for the sometimes dramatic differences in the samples’ efficacy were difficult to ascertain.²⁴

One of the chief problems was the difficulty of systematizing knowledge gathered from diverse localities. As the Joint Office emphasized, local factories had developed different manufacturing methods. In one factory, a plant might be pressed, while in another it might be soaked, boiled, or ground. This could have a significant impact on potency. They further noted that most of the materials were wild plants, and many of those were unfamiliar; even if the plants themselves were known to the scientists, the local names would likely be obscure.²⁵ This is a fundamental problem in translating between local and global knowledge systems; the rich profusion of local dialects in China made this a very familiar dilemma for scientists in many fields.

Academic work on native pesticides manifested most visibly in the encyclopedic handbooks known as *tunongyao zhi* (土农药志 records of native pesticides). These books typically adopted a strikingly practical tone: unlike many other kinds of books, they presented very little in the way of explicit

由茶麸 (茶麸, 茶籽麸皮)、一种在侗语中被称为果心叶 (果心叶, 果心叶) 的植物、灰烬和一种被称为“鸡血藤”的草药 (在其他地方被称为“鸡血藤”和许多其他名称) 制成; 经发现, 它对稻螟虫的防治有效率达95%。文章强调: “全国各地都有类似的例子。”²²

动员农村人口, 并在当地农药厂应用传统医学知识 (被认为是劳动人民几个世纪以来积累的智慧): 这是典型的土科学。但本土农药运动有望为“土洋结合”创造“物质基础”, 从而“为技术革命开辟新道路”。²³ 洋科学的介入在于系统地组织和传播地方经验, 对草药进行化学分析, 以及由专业研究人员对农药进行测试。

成立于1958年的地方农药科学研究联合办公室直接对农药及其靶标进行研究。为了不浪费冬春两季的时间, 他们在实验室里培育昆虫并测试农药——他们承认实验室条件与田间条件不同, 但仍坚持认为所产生的信息是有用的。他们测试了404种地方农药对蚊子幼虫的作用。在每种情况下, 他们都测试了用水和酒精制成的酊剂, 他们还比较了从不同地点收集的相同物种, 尽管样本功效有时存在显著差异的原因难以确定。²⁴

主要问题之一是难以将从不同地区收集的知识系统化。正如联合办公室所强调的, 地方工厂开发了不同的生产方法。在一家工厂, 植物可能被压榨, 而在另一家工厂, 它可能被浸泡、煮沸或研磨。这可能对效力产生重大影响。他们进一步指出, 大多数材料是野生植物, 其中许多是不熟悉的; 即使科学家们知道这些植物本身, 当地的名称也可能不为人知。²⁵ 这是在地方和全球知识体系之间进行转换的一个基本问题; 中国地方方言的丰富多样性使得这成为许多领域科学家非常熟悉的困境。

tunongyao zhi (土农药志, 即本土农药记录) 等百科全书式手册的出现, 最明显地体现了本土农药的学术研究。这些书籍通常采用引人注目的实用语调: 与许多其他类型的书籍不同, 它们很少明确地阐述理论。

ideology, confining the political discourse to a preface or two. However, the books nonetheless reflected Maoist ideology in their very makeup. Most significantly, the volumes drew on information gathered in different localities from around China and often specified in each case where information on the different herbs had come from; this exemplified the relationship between *yang* and *tu* and resonated more broadly with the Maoist philosophy of the mass line, in which the wisdom of the masses would be systematized to create more generalizable and extendable knowledge.²⁶

And yet, there were limits on just how earthy, how grassroots, how *tu*, published materials on native insecticides got. These limits are particularly noticeable in the faintness of the traces left by traditional medical knowledge—particularly the pattern of references to *xingwei*, to the eighteen conflicting medicines, and to the consultation of practitioners of traditional medicine. The prominent examples appear almost exclusively in materials relating to the campaign as it unfolded in Sichuan.²⁷ While it is possible that this reflects a regional strength in traditional Chinese medicine, I think it is more likely a result of how early the campaign work began in Sichuan, before much guidance had emerged from central authorities: it was thus more of a local effort, and officials were truly making do with local resources.

The afterword of a handbook published by Da County (Sichuan) in May 1959 strongly emphasized the importance of knowing the *xingwei* of the plants so that they could be combined effectively, but the descriptions of the plants in the main body of the text did not include this information. Similarly, the book published in Kaifeng in 1960 included information on *xingwei* in one table charting the properties of various plants; but a second table contained no information on this point—in that one, the information was copied from an authoritative guide produced by the Agricultural Ministry in September 1958, which contained no mention of *xingwei*, Chinese doctors, or anything else relevant to traditional Chinese medicine.²⁸ Tellingly, the most *tu* form that research on native pesticides took—drawing on knowledge of local plants passed down through the generations by medical practitioners and farmers—did not become inscribed in the most influential sources produced by central authorities.

Scholars have often seen the processes of systematization and rationalization as risking the erasure of local knowledge and flattening of traditional epistemologies. In contrast, Lili Lai and Judith Farquhar have offered a compellingly optimistic analysis of the state-sponsored systematizing of national

意识形态，将政治论述限制在一两个序言中。然而，这些书籍本身就反映了毛泽东思想。最重要的是，这些卷册借鉴了从全国各地收集的信息，并且通常在每种情况下都指明了不同草药信息的来源；这体现了阳与土之间的关系，并更广泛地与毛泽东的群众路线哲学产生共鸣，即群众的智慧将被系统化，以创造更具普遍性和可扩展性的知识。²⁶

然而，关于本土杀虫剂的出版物，其“土”的程度、草根的程度、土的程度都存在限制。这些限制在传统医学知识留下的微弱痕迹中尤为明显——特别是对性味、十八反药物以及咨询传统医学从业者的提及模式。突出的例子几乎只出现在与四川地区运动展开相关的材料中。²⁷虽然这可能反映了四川在中医药方面的区域优势，但我认为这更可能是由于四川的运动工作开始得较早，当时中央尚未出台太多指导：因此，这更多地是一项地方性工作，官员们确实在利用地方资源。

1959年5月大邑县（四川）出版的一本手册的后记，强烈强调了了解植物性味的重要性，以便有效结合，但正文对植物的描述中并未包含此信息。同样，1960年开封出版的书中，在一张列出各种植物特性的表格中包含了性味信息；但第二张表格中却没有这方面的信息——那张表格的信息是从农业部1958年9月发布的一份权威指南中复制的，其中没有提及性味、中医或任何与中医药相关的内容。²⁸值得注意的是，对本土农药研究最土的形式——借鉴世代相传的医者和农民的本地植物知识——并未被中央当局最具影响力的资料所记载。

学者们通常认为，系统化和合理化的过程可能会抹去地方知识，并使传统认识论趋于扁平化。与此相反，赖丽丽和 Judith Farquhar 对国家资助的系统化工作提出了令人信服的乐观分析

minority medicines in contemporary China. In their view, the “rationalizing process” that “has emerged as a modernizing imperative” has simultaneously “generated new charismatic forms of healing effectiveness and authority.”²⁹ As we will see in more detail in the following section, the systematizing of *tu* plant knowledge in the native pesticides campaign could be read as generating “new charismatic forms”; still, the erasure of local actors bears watching.

GARLIC

A concrete example will afford a more tangible sense of how the handbooks worked in practice. *Dasuan* (大蒜 garlic *Allium sativum*) has occupied an important place in Chinese food culture since it was imported from western Asia during the Han dynasty.³⁰ Its medicinal properties were also recognized then, and by the time Li Shizhen compiled his *Bencao gangmu* (本草纲目 *Compendium of Materia Medica*, 1578), garlic had acquired numerous applications, from snake bites to warm-factor disorders. Consistent with this foundation, in handbooks on native pesticides, garlic was commonly recommended to treat rice blast (or “rice heat disease”).

The Ministry of Agriculture’s September 1958 book had this to say in its entry on garlic:

A member of the lily family. Its bulb and leaves have a very strong odor. Can be used as medicine.

In Guangxi they take 3 *jin* of peeled garlic bulbs, mash it into a paste, and put it in 100 *jin* of clear water to soak for half an hour to control *dao rebing* (稻热病 rice blast). After spraying 1–2 times, rice blast will decrease by 30–50 percent. It is more effective than Bordeaux [a common fungicide mixture made from copper sulfate, lime, and water] and is also effective against brown spot disease. In Anhui, they mash one *jin* of garlic and soak it in 10 *jin* of water to control stem rust in rice.

In Taizhou, Zhejiang, they have experimented with taking 5 *jin* of mashed garlic bulb and soaking it in 100 *jin* of water to control rice blast, with very good results. In Cixi, Zhejiang, Comrade Li Minghua of the Bureau of Agriculture has conducted preliminary experiments taking 100 *jin* of mashed garlic and adding 20 *jin* of water to control angular leaf spot in cotton, with results approximating those for chemical sprays.

In Changshu, Jiangsu, they squeeze half a *jin* of liquid from 1 *jin* of garlic, then add 3 *liang* (150 g.) of camphor for every 1 *jin* of garlic juice. After mixing thoroughly, they soak it in 4 *jin* of water. In controlling cotton aphids it is 82.2 percent effective.

中国当代少数民族医药。在他们看来，“作为现代化当务之急而出现的‘合理化进程’”同时“产生了新的魅力型治疗效果和权威形式”。²⁹正如我们将在下一节中更详细地看到，在本土农药运动中，对土植物知识的系统化可以被解读为产生了“新的魅力形式”；然而，对地方行为者的抹杀值得关注。

大蒜

一个具体的例子将更直观地说明这些手册在实践中是如何运作的。大蒜（大蒜 *Allium sativum*）自汉代从西亚传入以来，在中国饮食文化中一直占据重要地位。³⁰其药用价值当时也得到了认可，到李时珍编纂《本草纲目》（本草纲目，1578年）时，大蒜已有了多种用途，从蛇咬伤到温热病症。与此基础一致，在本土农药手册中，大蒜常被推荐用于治疗稻瘟病（或“稻热病”）。

农业部1958年9月的书在其大蒜条目中写道：

百科植物。其鳞茎和叶子有很强的气味。可药用。

在广西，他们取3斤去皮蒜瓣，捣成糊状，放入100斤清水中浸泡半小时，以防治稻热病（稻热病）。喷洒1-2次后，稻热病可减少30-50%。它比波尔多液[一种由硫酸铜、石灰和水制成的常见杀菌剂混合物]更有效，对褐斑病也有效。在安徽，他们将一斤大蒜捣碎，浸泡在10斤水中，以防治水稻秆锈病。

在浙江台州，他们曾试验用5斤捣碎的蒜瓣浸泡在100斤水中来防治稻瘟病，效果非常好。在浙江慈溪，农业局的李明华同志曾初步试验用100斤捣碎的大蒜加入20斤水来防治棉花角斑病，效果接近化学喷雾剂。

在江苏常熟，他们从1斤大蒜中榨出半斤液体，然后每1斤大蒜汁中加入3两（150克）樟脑。充分混合后，将其浸泡在4斤水中。在防治棉蚜方面，其有效率为82.2%。

In contrast, a provincial-level volume published by the Fujian Agricultural Tool Reform Office in November 1958 included a longer description of the plant itself and an illustration, but no discussion of practices from other places.

Some handbooks, including the handwritten one produced in Kaifeng mentioned in the introduction, copied the Agricultural Ministry's entries word for word; the meticulous hand-copying of the Kaifeng volume suggests that the Ministry's handbook was not available in sufficient numbers, such that redistribution even in this laborious way using scarce fiber resources was considered valuable.

When the Joint Office for Scientific Research on Native Pesticides published its eagerly awaited volume in May 1959, it offered somewhat longer entries for each plant, including different sections on morphological characteristics, distribution, growing environment, composition (complete with chemical formulas for the effective agents), uses, and preparation methods and targets of control. In the last section of the garlic entry, it listed eight methods and targets, most of which were attributed to specific provinces or counties. This format was chosen for imitation by some other handbooks, for example the one published in Da County, Sichuan.

Both the similarities and the differences among these materials deserve notice. Most obviously, the books shared the common purpose of providing accessible information necessary for the production of native pesticides. Some presented detailed descriptions of the plants' morphology, a characteristic of modern field guides not shared with traditional medicinal encyclopedias like Li Shizhen's Ming-Dynasty *Bencao gangmu*. In other ways, some of the handbooks are quite reminiscent of those older texts—for example, in the organization of each entry into categories of information and in the listing of documented applications, in each case attributed to a specific source. Though many borrowed liberally from one another, these were by no means simple reproductions. Each individual text involved efforts to tailor available information to the perceived needs of local users, and each also involved decisions made by the individual authors and editors assembling the content regarding the most user-friendly format and most appropriate language.

In 1960, the journal *Chemistry World* reported on experiments isolating the chemical components found in garlic and demonstrating their antimicrobial effects in treating crop diseases. According to the authors, by moving

相比之下,福建省农业工具改革办公室于1958年11月出版的一本省级刊物,对植物本身进行了更长的描述和插图,但没有讨论其他地方的实践。

一些手册,包括引言中提到的开封手抄本,逐字逐句地复制了农业部的条目;开封卷的细致手抄表明农业部的手册数量不足,以至于即使以这种费力的方式,利用稀缺的纤维资源进行再分发也被认为是很有价值的。

1959年5月,全国土农药科学研究联合办公室出版了其备受期待的著作,其中对每种植物的介绍都更长,包括形态特征、分布、生长环境、成分(附有有效成分的化学式)、用途以及制备方法和防治目标等不同部分。在大蒜条目的最后一部分,它列出了八种方法和目标,其中大部分归因于特定的省份或县。这种格式被其他一些手册效仿,例如四川大县出版的手册。

这些材料的相似之处和不同之处都值得注意。最明显的是,这些书籍都有一个共同的目的,即提供生产本地农药所需的易于获取的信息。有些书籍详细描述了植物的形态,这是现代野外指南的特点,而李时珍明代《本草纲目》等传统医药百科全书则不具备这一特点。在其他方面,一些手册与那些较早的文本非常相似——例如,每条目都按信息类别组织,并列有有据可查的应用,每种情况都归因于特定来源。尽管许多书籍相互借鉴,但它们绝不是简单的复制品。每部独立的文本都努力根据当地用户的感知需求调整现有信息,每部文本还涉及个人作者和编辑在组装内容时就最用户友好的格式和最合适的语言所做的决定。

1960年,《化学世界》杂志报道了分离大蒜中化学成分并证明其在治疗作物病害方面的抗菌作用的实验。据作者称,通过移动

from the plant itself to the chemicals it contains, the research fulfilled a key goal of the era: “*tuzhong chuyang*” (土中出洋 produce the *yang* from out of the *tu*). That is, with simple and locally available ingredients, create substitutes for expensive materials that China would otherwise be dependent on foreign sources to supply.³¹ While the more common slogan remained “combine *tu* and *yang*,” the sense that *tu* was a steppingstone (needs must, faute de mieux) to *yang* is palpable in many of the published materials. This helps explain the evidence the handbooks supply as to the integration of *tu* and *yang*: while we find conscious efforts to retain traces of the origins of local knowledge, there is at least as much deliberate incorporation of both the trappings and substance of professional science. And we see the ways in which certain aspects of *tu* science—in particular, the contributions of traditional doctors, monks, and other suspect actors—were erased as the knowledge became systematized and generalized in these texts.

POISON

Like so much else about the Great Leap Forward, the campaign for native pesticides began with optimistic fanfare in 1958 but became ever more complex and dissonant over the subsequent years. Nowhere is this more apparent than in the many accounts of poisonings, when desperate people turned to eating noxious wild plants, including some used as botanical pesticides.

The Great Leap-era sources exhibit a tension between the ostensibly natural and wholesome character of native pesticides and their powerful toxicity. Sources frequently pointed to the safety of botanical pesticides and their beneficial effects on soil health and fertility. As one article explained, the safety of botanicals was especially significant when treating fruits and vegetables that are eaten as is—with foreign pesticides, harvest would have to wait one or two weeks before harvesting and eating, while crops treated with native pesticides could be harvested and consumed immediately.³² And yet, Great Leap-era sources often highlighted the risks involved in manufacturing native pesticides. While some articles made sanguine references to local people smelling and tasting the herbs to determine their *xingwei*, others specifically warned against haphazardly smelling or tasting and encouraged people to wear a mask and goggles when working with the plants—and even to place masks on livestock when processing the pesticides. They also sometimes offered first-aid recommendations in case of accidental poisoning: for

从植物本身到其所含的化学物质，这项研究实现了一个时代的关键目标：“土中出洋”（土中出洋，从土中生产出洋）。也就是说，用简单和本地可得的原料，为中国原本依赖外国供应的昂贵材料创造替代品。³¹虽然更常见的口号仍然是“土洋结合”，但在许多已出版的材料中，土是通向洋的垫脚石（权宜之计，不得已而为之）的感觉是显而易见的。这有助于解释手册中提供的关于土和洋整合的证据：虽然我们发现有意识地努力保留地方知识起源的痕迹，但至少同样多的是有意识地融入了专业科学的形式和实质。我们还看到，随着知识在这些文本中系统化和普遍化，土科学的某些方面——特别是传统医生、僧侣和其他可疑人物的贡献——是如何被抹去的。

毒药

与大跃进的许多其他方面一样，本土农药运动始于1958年乐观的宣传，但在随后的几年中变得越来越复杂和不协调。这一点在许多中毒事件的记载中表现得尤为明显，当时绝望的人们转而食用有毒的野生植物，其中包括一些用作植物性农药的植物。

大跃进时期的资料显示，本土农药表面上天然、有益健康的特性与其强大的毒性之间存在张力。资料经常指出植物性农药的安全性及其对土壤健康和肥力的有益影响。正如一篇文章所解释的，植物性农药的安全性在处理即食水果和蔬菜时尤为重要——使用外国农药，收获后需要等待一到两周才能食用，而用本土农药处理的作物可以立即收获和食用。³²然而，大跃进时期的资料也经常强调制造本土农药所涉及的风险。虽然有些文章乐观地提到当地人通过闻和尝草药来确定其性味，但另一些文章则明确警告不要随意闻或尝，并鼓励人们在处理植物时佩戴口罩和护目镜——甚至在加工农药时给牲畜戴上口罩。他们有时还会提供意外中毒时的急救建议：例如

example, if someone were poisoned with a plant classified as numbing, they should use a plant classified as spicy to rinse out their mouths and follow by ingesting rice or mung beans.³³

Poisonings sometimes occurred for other, more tragic reasons. One was the ingestion of toxic plants in an effort to consume what the state referred to as *dai shipin* (代食品 food substitutes). According to PRC historian Gao Hua, this term was first used in 1955 when peasants in Guangxi Province collected wild fruit and tree bark to eat, in place of the grain that had been taken by the state under the Unified Purchase and Sale Policy. In 1960, in the face of increasingly impossible-to-ignore reports of food shortages, Mao called for more food to be grown and more so-called food substitutes to be found. Commonly promoted food substitutes included the algae *chlorella*; artificial starch made by grinding stalks, roots, and other plant parts into a powder to be added to dough for steamed buns; and artificial meat produced with bacteria cultures. Perhaps ironically, the Insect Research Institute of the Chinese Academy of Sciences, which was more often concerned with the protection of crops from insect pests, recommended collecting and eating insects to bolster nutrition.³⁴

The increasingly desperate search for food substitutes also led to alarming rises in cases of accidental poisoning. Sometimes the plants in question were in fact food crops but required special processing to remove toxins. Such was the case with cassava, which is the source of tapioca starch but contains high quantities of cyanide in its raw form: a telephone report from Fujian Province to the central authorities in December 1960 blamed cassava for poisoning 2,071 people with 294 fatalities: presumably, desperate hunger led people to consume the plant raw.³⁵

While cassava was not to my knowledge used as a pesticide, other plants blamed for poisonings were. Three such plants were the *huaishu* (槐树 scholar tree *Sophora japonica*), *chunshu* or *chouchun* (椿树 or 臭椿 tree of heaven *Ailanthus altissima*), and *cang'er* (苍耳 Siberian cocklebur *Xanthium strumarium*).³⁶ A report on the famine from Shandong Province in January 1959 documented that after consumption of scholar tree and tree of heaven, “the entire body would become swollen.” In the worst instances, “the body could no longer retain excess fluid, which then erupted through the skin, oozing out with a yellowish color.”³⁷ In April 1960, following reports of numerous poisonings from cocklebur seed, the Ministry of Health attempted to ban its promotion—the promotion, and the poisonings, continued nonetheless.³⁸

例如，如果有人被归类为麻木的植物毒害，他们应该使用归类为辛辣的植物漱口，然后食用米饭或绿豆。³³

中毒有时是出于其他更悲惨的原因。其中之一是摄入有毒植物，以食用国家所称的代食品（代食品食物替代品）。据中华人民共和国历史学家高华称，这个词最早在1955年使用，当时广西省的农民收集野果和树皮来吃，以代替国家根据统购统销政策征收的粮食。1960年，面对日益无法忽视的粮食短缺报告，毛泽东呼吁种植更多粮食，并寻找更多所谓的食物替代品。普遍推广的食物替代品包括小球藻；通过将茎、根和其他植物部分磨成粉末添加到馒头面团中制成的人造淀粉；以及用细菌培养物生产的人造肉。或许具有讽刺意味的是，中国科学院昆虫研究所（该研究所更常关注保护作物免受害虫侵害）建议收集和食用昆虫以补充营养。³⁴

人们对食物替代品日益绝望的寻找也导致意外中毒病例的惊人增加。有时，所涉植物实际上是粮食作物，但需要特殊加工才能去除毒素。木薯就是这种情况，它是木薯淀粉的来源，但生食时含有大量氰化物：1960年12月，福建省向中央当局报告称，木薯导致2071人中毒，294人死亡：据推测，绝望的饥饿导致人们生食这种植物。³⁵

据我所知，木薯并未用作杀虫剂，但其他被指控导致中毒的植物却被用作杀虫剂。其中三种植物是槐树（槐树，国槐 *Sophora japonica*）、椿树或臭椿（椿树或臭椿，臭椿 *Ailanthus altissima*）和苍耳（苍耳，苍耳 *Xanthium strumarium*）。³⁶1959年1月，山东省一份关于饥荒的报告记载，食用槐树和臭椿后，“全身都会肿胀”。在最严重的情况下，“身体无法再保留多余的液体，然后液体会从皮肤中喷出，渗出黄色液体”。³⁷1960年4月，在接到大量苍耳子中毒报告后，卫生部试图禁止其推广——然而，推广和中毒事件仍在继续。³⁸

The horror of famine also led people to suicide, and intentionally ingesting pesticide was one frequently cited method.³⁹ It is unclear whether any of these cases involved native pesticides; the greater potency of chemical pesticides would make them far more likely resources for this ghastly endeavor of last resort. But the same principle of scarcity that propelled the native pesticides campaign in the first place may have led some to try botanical alternatives, especially if they were readily available in a local factory.

These are all horrible fates to contemplate, and nothing lightens the tragedy. At the same time, it is reasonable to expect that the number of accidental poisonings would have been considerably greater if not for the knowledge about wild plants that was widely held in rural communities—as documented especially in the early stages of the native pesticide campaign, before *yang* knowledge had caught up with *tu* knowledge. Where people resorted to eating poisonous plants, it was probably after the edible plants had already been consumed. Where the shortages did not quite reach that stage, many people no doubt survived because they knew which parts of which plants were poisonous and which were safe to eat.

LEGACY

Recent years have seen a surge of interest in what is now being called *Zhongyi nongye* (中医农业 Chinese medicine agriculture).⁴⁰ In 2016, I met with a proponent, Zhu Weiping, and visited two of the experimental sites he established with the organic agriculture company that he helped run. He had not heard of the Great Leap-era native pesticide campaign, nor did he seem inclined to explore the connection. This did not surprise me. The tragedy of the Great Leap famine and the more general reputation of the Great Leap as a colossal failure make it an unlikely source of inspiration for hopeful scientists and entrepreneurs in China today. And yet, there is no question that the Chinese medicine agriculture of the present belongs to the same historical narrative as the campaign we have examined in these pages. Indeed, it has far more in common with the Great Leap campaign than with the imperial-era history it is more inclined to evoke. The richest sources produced on “traditional” knowledge of botanical insecticides were produced in the Great Leap Forward, while only scattered records exist for the imperial era.

A key difference between today’s Chinese medicine agriculture and the native pesticides campaign of the Great Leap Forward lies in the relationship

饥荒的恐怖也导致人们自杀，故意摄入农药是经常被提及的一种方法。³⁹ 目前尚不清楚这些案例中是否涉及本土农药；化学农药的效力更强，使其更有可能成为这种可怕的最后手段。但正是最初推动本土农药运动的稀缺性原则，可能导致一些人尝试植物替代品，特别是如果它们在当地工厂很容易获得的话。

这些都是令人不寒而栗的命运，没有任何事物能减轻这场悲剧。与此同时，可以合理地预期，如果不是农村社区普遍掌握的野生植物知识——尤其是在本土农药运动早期，在阳知识赶上土知识之前——意外中毒的数量会大大增加。人们诉诸食用有毒植物，很可能是在可食用植物已被消耗殆尽之后。在短缺尚未达到那种程度的地方，许多人无疑幸存下来，因为他们知道哪些植物的哪些部分有毒，哪些可以安全食用。

传承

近年来，人们对现在被称为中医药农业（中医农业）的领域兴趣大增。⁴⁰ 2016年，我拜访了倡导者朱维平，并参观了他与他帮助经营的有机农业公司建立的两个实验基地。他没有听说过大跃进时期的土农药运动，似乎也不倾向于探究二者之间的联系。这并不令我感到惊讶。大跃进饥荒的悲剧以及大跃进作为一场巨大失败的普遍声誉，使其不太可能成为当今中国充满希望的科学家和企业家的灵感来源。然而，毫无疑问，当今的中医农业与我们在这些篇章中探讨的运动属于同一历史叙事。事实上，它与大跃进运动的共同点远多于它更倾向于唤起的帝国时代历史。关于植物杀虫剂“传统”知识的最丰富资料是在大跃进时期产生的，而帝国时代只有零星的记载。

当今中医药农业与大跃进时期“土农药”运动的一个关键区别在于二者之间的关系

of each episode to industrialized agriculture. In 1958, the PRC sought to transform Chinese agriculture as part of the larger goal of creating a modern, industrial nation. The lack of chemical pesticides presented one of many obstacles related to the overarching conundrum of scarcity; here, as in so many other areas, Chinese farmers, workers, and officials were urged to embrace “native” materials and knowledge—at times characterized as a great “treasure-house” and at other times (perhaps more honestly, given overwhelming evidence of the nation-state’s modernizing priorities) as merely a stop-gap measure. And, of course, during the famine the scarcity of chemicals paled in comparison with the scarcity of food itself: so scarce that algae and acorns were promoted as substitutes, and people resorted to tree bark and wild plants when nothing else was available.

Today, industrialized agriculture dominates the Chinese landscape, and chemical pesticides have, since the 1970s, ceased being a scarce resource and have instead increasingly become a happy abundance or a scourge, depending on one’s point of view. The environmental and health consequences of this transformation have inspired an impressive array of movements for more “natural” food production—fueled in large part by the increasingly abundant disposable income of the urban middle and upper classes. That these movements include not only “organic” and “ecological” principles, but also the “traditional” principles of Chinese medicine, speaks to the commercial value of heritage in contemporary China—reflected also in the marketing of “heritage” breeds of meat animals and “heritage” varieties of grains and vegetables.

The native pesticides campaign of the Great Leap Forward and present-day alternative food movements thus occupy different positions within the larger historical arc of agricultural modernization in China. Throughout this history, we see state agents, scientists, and rural people actively mobilizing diverse forms of knowledge in their efforts to find nourishment and healing for people and their crops. The stark differences between the eras should not prevent us from recognizing the complexity of the challenges that Mao-era historical actors faced, the sophistication of their responses, or the familiarity of many of their fundamental concerns.

NOTES

1. Kaifengshi kexue jishu weiyuanhui and Kaifengshi kexuejishu xiehui, eds., *Tunong-fei tunongyao ziliao huibian* [Compilation of resources on native fertilizers and native pesticides] (no publishing info, 1960).

每一集都致力于工业化农业。1958年，中华人民共和国试图改造中国农业，作为创建现代化工业国家这一更大目标的一部分。化学农药的缺乏是与稀缺性这一首要难题相关的众多障碍之一；在这里，正如在许多其他领域一样，中国农民、工人和官员被敦促接受“本土”材料和知识——有时被描述为一个巨大的“宝库”，而有时（考虑到国家现代化优先事项的压倒性证据，或许更诚实地说）仅仅是一种权宜之计。当然，在饥荒期间，化学品的稀缺性与食物本身的稀缺性相比显得微不足道：食物稀缺到藻类和橡子都被推广为替代品，当没有其他东西可用时，人们只能求助于树皮和野生植物。

如今，工业化农业在中国占据主导地位，自20世纪70年代以来，化学农药已不再是稀缺资源，反而日益成为一种令人欣喜的丰富资源或一种祸害，这取决于人们的观点。这种转变对环境与健康造成的影响，激发了一系列令人瞩目的运动，旨在推动更“自然”的食品生产——这在很大程度上得益于城市中产阶级和上层阶级日益充裕的可支配收入。这些运动不仅包括“有机”和“生态”原则，还包括中医的“传统”原则，这表明了传统文化在当代中国的商业价值——这也体现在“传统”肉类动物品种以及“传统”谷物和蔬菜品种的营销中。

因此，大跃进时期的本土农药运动与当今的替代食品运动在中国农业现代化的大历史进程中占据着不同的位置。纵观这段历史，我们看到国家机构、科学家和农村人民积极调动各种知识，努力为人类和农作物寻找营养和治疗方法。时代之间的巨大差异不应妨碍我们认识到毛泽东时代的历史参与者所面临挑战的复杂性、他们应对措施的精妙之处，以及他们许多基本关切的熟悉性。

注释

1. 开封市科学技术委员会和开封市科学技术协会编：《土农—非土农药资料汇编》[《本地肥料和本地农药资源汇编》]（无出版信息，1960年）。

2. Jacob Eyferth, *Eating Rice from Bamboo Roots: The Social History of a Community of Handicraft Papermakers in Rural Sichuan, 1920–2000* (Cambridge, MA: Harvard University Asia Center, 2009).
3. Jakob A. Klein, “‘For Eating, It’s Guangzhou’: Regional Culinary Traditions and Chinese Socialism,” in *Enduring Socialism: Explorations of Revolution and Transformation, Restoration and Continuation*, ed. Harry G. West and Parvathi Raman (New York: Berghahn Books, 2009), 46.
4. Hanchao Lu, “The Tastes of Chairman Mao: The Quotidian as Statecraft in the Great Leap Forward and Its Aftermath,” *Modern China* 41, no. 5 (September 2015): 539–572, 543.
5. Qiao Gongjian and Yin Zutang, “Shinian lai yesheng zhiwu liyong fangmian de chengjiu” [Achievements in the last ten years related to the use of wild plants], *Shengwuxue tongbao* 1959, no. 10: 443; Hebeisheng weishengting liangshi tinghe, ed., *Yecai he daishipin* [Wild plants and food substitutes] (no publishing info, preface dated 1960).
6. Benkan bianweihui, “Kunchongxue zai nongye fengchan zhong de juda chengjiu” [Great achievements of entomology in producing bumper harvests], *Kunchong zhishi* 1958, no. 5: 199–200.
7. Lü Youlan, “Zongjie tuiguang tunongyao de shida haochu” [Ten benefits of extending native pesticides], *Kunchong zhishi* 1959, no. 1: 5–6.
8. Tunongyao kexue yanjiu lianhe bangongshi, “Dagao tunongyao de chengjiu ji jinhou fazhan” [Achievements and future development of the going big in native pesticides {campaign}], *Kunchong zhishi* 1958, no. 6: 251–252.
9. Sichuan Daxian zhuanqu kexue yanjiusuo, ed., *Tunongyao zhi* [Records of native pesticides] (Sichuansheng difang guoying Daxian yinshua chang, 1959), 190.
10. Sigrid Schmalzer, *Red Revolution, Green Revolution: Scientific Farming in Socialist China* (Chicago: University of Chicago Press, 2016); see also Jia-Chen Fu, “Artemisinin and Chinese Medicine as Tu Science,” *Endeavor* 41, no. 3 (2017): 127–135.
11. Marta E. Hanson, *Speaking of Epidemics in Chinese Medicine: Disease and the Geographic Imagination in Late Imperial China* (New York: Routledge, 2011), 1, 16, 171n3.
12. Wang Qizhong and Liu Shengming, “Sichuan Daxian dagao tunongyao de jingyan” [Experiences from Da County, Sichuan, in going big in native pesticides] *Kunchong zhishi* 1959, no. 1: 29.
13. Lü Youlan, “Zongjie tuiguang.”
14. Yue Zong, “Zhongguo tunongyao lishi ziliao zhailu” [Excerpts from historical materials on Chinese native pesticides], *Kunchong zhishi* 1959, no. 1: 37.
15. Tunongyao kexue, “Dagao tunongyao.”
16. Joseph Needham with Lu Gwei-djen and Huang Hsing-tsung, *Botany*, in *Science and Civilisation in China*, vol. 6: Biology and Biological Technology, pt. 1 (Cambridge: Cambridge University Press, 1986), 471–553.

2. 雅各布·艾弗斯, 《食竹根米: 四川农村手工造纸社区的社会史, 1920-2000》(马萨诸塞州剑桥: 哈佛大学亚洲中心, 2009)。
3. 雅各布·A·克莱因, “‘吃在广州’: 区域烹饪传统与中国社会主义”, 载于哈里·G·韦斯特和帕瓦蒂·拉曼编著的《持久的社会主义: 革命与转型、复辟与延续的探索》(纽约: 博格汉出版社, 2009), 第46页。
4. 卢汉超, “毛主席的口味: 大跃进及其余波中的日常生活与治国之道”, 《现代中国》41卷第5期(2015年9月): 第539-572页, 第543页。
5. 乔公健和殷祖堂, “十年野生植物利用方面的成就” [Achievements in the last ten years related to the use of wild plants], 生物学通报 1959年, 第10期: 443页; 河北省卫生厅粮食厅和编, 野菜和代食品 [Wild plants and food substitutes] (无出版信息, 序言日期为1960年)。
6. 本刊编委会, “昆虫学在农业丰产中的巨大成就” [Great achievements of entomology in producing bumper harvests], 昆虫知识1958年, 第5期: 199-200页。
7. 吕友兰, “总结推广土农药的十大好处” [Ten benefits of extending native pesticides], 昆虫知识 1959年, 第1期: 5-6页。
8. 桐农药科学研究联合办公室, 《大搞桐农药的成就及今后发展》[土农药大发展的成就和未来发展{campaign}], 昆虫知识 1958年第6期: 251-252页。
9. 四川达县专区科学研究所编, 《土农药志》[土农药记录] (四川省地方国营达县印刷厂, 1959年), 第190页。
10. Sigrid Schmalzer, 《红色革命, 绿色革命: 社会主义中国的科学农业》(芝加哥: 芝加哥大学出版社, 2016); 另见贾晨甫, “青蒿素与中医作为土科学”, 《奋进》41卷第3期(2017): 127-135页。
11. Marta E. Hanson, *Speaking of Epidemics in Chinese Medicine: Disease and the Geographic Imagination in Late Imperial China* (New York: Routledge, 2011), 1, 16, 171n3.
12. 王启忠、刘盛明, “四川大县大搞土农药的经验” [Experiences from Da County, Sichuan, in going big in native pesticides] 昆虫知识 1959, no. 1: 29.
13. 吕友兰, “总结推广。”
14. 岳宗, “中国土农药历史资料摘录” [Excerpts from historical materials on Chinese native pesticides], 昆虫知识 1959, no. 1: 37.
15. 土农药科学, “大搞土农药。”
16. Joseph Needham with Lu Gwei-djen and Huang Hsing-tsung, *Botany*, 载于 *Science and Civilisation in China*, 第6卷: Biology and Biological Technology, 第1部分 (剑桥: 剑桥大学出版社, 1986), 第471-553页。

17. Lü Youlan, “Zongjie tuiguang.”
18. Sichuan Daxian shangyeju, “Daxian tunongyao zhizao, shiyong, chucun bao-guan fangfa” [Methods of manufacturing, using, and safely storing native pesticides in Da County], *Nongye kexue tongxun* 1959, no. 12: 421–422.
19. Sichuan Daxian zhuanqu kexue yanjiusuo, *Tunongyao zhi*, 191.
20. Sichuan Daxian shangyeju, “Daxian tunongyao.”
21. Lü Youlan, “Zongjie tuiguang.”
22. Benkan weiyuanhui, “Kunchongxue.”
23. Lü Youlan, “Zongjie tuiguang.”
24. Zhongguo kexueyuan kunchong yanjiusuo huaxue fangzhi shi, “1958 nian quan-guo tunongyao yaoxiao ceding chubu baogao” [Preliminary report on determining the efficacy of native pesticides throughout the nation in 1958], *Kunchong zhishi* 1959, no. 1: 4, 7–16; Zhongguo nongye kexueyuan zhiwu baohu yanjiusuo, “Tunongyao dui zhuyao nongzuowu bingchong de yaoxiao ceding” [Determining the efficacy of native pesticides in important crop diseases and insect pests], *Nongye kexue tongxun* 1959, no. 9: 297–300.
25. Tunongyao kexue, “Dagao tunongyao.”
26. See, for example, Nongye ziliao bianji weiyuanhui, ed., *Tunongyao* [Native pesti-cides] (Beijing: Nongye chubanshe, 1958).
27. That said, an article on the native pesticide campaign from August 1958 in Jiangxi Daily made passing reference to the consultation of traditional doctors, sug-gesting that the practice was not limited to Sichuan. Reprinted in in Nongye ziliao, *Tunongyao*, 16.
28. Kaifengshi, *Tunongfei*, 54–58.
29. Lili Lai and Judith Farquhar, “Nationality Medicines in China: Institutional Ratio-nality and Healing Charisma,” *Comparative Studies in Society and History* 57.2 (2015): 381–406, 384.
30. Wang Saishi, “Gudai dui dasuan de yinjin yu liyong” [Introduction and use of garlic in ancient times], *Nongye kaogu* 1996, no. 1: 182–188.
31. Hai Binfu, “Yizhong baogui tunongyao dasuan de huaxue chengfen” [Chemical composition of garlic, a precious native pesticide], *Huaxue shijie* 1960, no. 2: 56–57.
32. Lü Youlan, “Zongjie tuiguang,” 6.
33. Sichuan Daxian shangyeju, “Daxian tunongyao.” See also Wang and Liu, “Sichuan Daxian,” 31.
34. Gao Hua, “Food Augmentation Methods and Food Substitutes during the Great Famine,” in *Eating Bitterness: New Perspectives on China’s Great Leap Forward and Famine*, eds. Kimberly Ens Manning and Felix Wemheuer (Vancouver: University of British Columbia Press, 2011), 183–187.

17. 吕幼兰, “总结推广。”
18. 四川大县商业局, “大县土农药制造、使用、贮存保管方法” [大县土农药的制造、使用和安全储存方法], 农业科学通讯 1959年第12期: 421-422。
19. 四川大县专区科学研究所, 土农药志, 191。
20. 四川大县商业局, “大县土农药。”
21. 吕友兰, “总结推广。”
22. 本刊委员会, “昆虫学。”
23. Lü Youlan, “Zongjie tuiguang.”
24. 中国科学院昆虫研究所化学防治室, “1958年全国土农药药效测定初步报告” [Preliminary report on determining the efficacy of native pesticides t hroughout the nation in 1958], 昆虫知识 1959, no. 1: 4, 7–16; 中国农业科学院植物保 护研究所, “土农药对主要农作物病虫的药效测定” [Determining the efficacy of native pesticides in impor tant crop diseases and insect pests], 农业科学通讯 1959, no. 9: 297–300。
25. 土农药科学, “大搞土农药。”
26. 例如, 参见农业资料编辑委员会编: 《土农药》[Native pesti- cides] (北京: 农业出版社, 1958)。
27. 尽管如此, 1958年8月《江西日报》一篇关于土农药运动的文章顺带提到了咨询中医, 这表明这种做法并非仅限于四川。转载于《农业资料》土农药, 第16页。
28. 开封市, 《土农肥》, 54-58页。
29. Lili Lai和Judith Farquhar, “Nationality Medicines in China: Institutional Ratio- nality and Healing Charisma”, 《比较社会与历史研究》57.2 (2015): 381– 406, 384。
30. 王赛时, “古代对大蒜的引进与利用” [Introduction and use of garlic in ancient times], 《农业考古》1996年第1期: 182-188页。
31. 海滨福, “一种宝贵土农药大蒜的化学成分” [大蒜的化学成分, 一种珍贵的本土农 药], 化学世界 1960年, 第2期: 56-57页。
32. 吕友兰, “总结推广”, 6页。
33. 四川达县商业局, “达县土农药”。另见王和刘, “四川达县”, 31页。
34. 高华, “大饥荒时期的食物增补方法和食物替代品”, 载于金伯利·恩斯·曼宁和费利 克斯·温赫尔编著的《吃苦: 中国大跃进和大饥荒的新视角》(温哥华: 不列颠哥伦比亚 大学出版社, 2011年), 183-187页。

35. Xun Zhou, *Great Famine in China, 1958–1962: A Documentary History* (New Haven, CT: Yale University Press, 2012), 51.
36. Sources promoting the use of these plants as insecticides include: Zhongguo renmin gongheguo nongyebu, ed., *Zhongguo tunongyao zhi* [Records of Chinese native pesticides] (Kexue chubanshe, 1959); Nongyebu, *Tunongyao zhi*; “Lixing jieyue, wei zuguo: wei zuguo chuanguang gengduo de caifu” [Be parsimonious to create more wealth for China], *Renmin ribao*, December 9, 1954, 2; and Yang Min, “Nongyao shihua,” *Renmin ribao*, April 3, 1960, 8.
37. Xun Zhou, *Great Famine*, 5.
38. Gao Hua, “Food,” 188–189; “Duo diao xie gongye yuanliao jincheng you shenme haochu” [Benefits of moving more industrial materials into the cities], *Renmin ribao*, October 27, 1961, 3.
39. See Xun Zhou, *Great Famine*, 8, among many other anecdotes.
40. See, for example, Zhang Lijian and Zhi Lizhi, “‘Zhongyi nongye’ fazhan zhanlue ji qianjing” [Strategies and prospects for the development of “Chinese medicine agriculture”], *Nongyeshengchan zhanwang* 2018, no. 11: 72–76.

35. 周逊, 《中国大饥荒, 1958–1962: 文献史》(纽黑文, 康涅狄格州: 耶鲁大学出版社, 2012), 第51页。
36. 推广使用这些植物作为杀虫剂的资料包括: 中华人民共和国农业部编, 《中国土农药志》[Records of Chinese native pesticides] (科学出版社, 1959); 农业部, 《土农药志》; “厉行节约, 为祖国: 为祖国创造更多的财富” [Be parsimonious to create more wealth for China], 《人民日报》, 1954年12月9日, 第2版; 以及杨敏, “农药史话”, 《人民日报》, 1960年4月3日, 第8版。
37. 周逊, 《大饥荒》, 第5页。
38. 高华, “食物”, 第188-189页; “多调些工业原料进城有什么好处” [Benefits of moving more industrial materials into the cities], 《人民日报》, 1961年10月27日, 第3页。
39. 参见周逊, 《大饥荒》, 第8页, 以及许多其他轶事。
40. 例如, 参见张立建和智利志, “‘中药农业’发展战略及前景” [Strategies and prospects for the development of “Chinese medicine agriculture”], 《农业生产展望》2018年第11期: 第72-76页。

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